



PreTUPS 5.1

External System Interface API (C2S)

Version: 1.1

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DOCUMENT HISTORY

Doc version	Version date	Modification
1.0	10/11/2010	Initial Version Release
1.1	29/06/2011	

Document Information

1.1 Introduction

This document describes the external interface between the Prepaid Top-up Server (PreTUPS) and an External system. External system can be any system, which can send XML over Http based request to PreTUPS. This interface provides the capability for an end subscriber to be able to Top-up their mobile prepaid account by one of the following methods:

- ♦ By using an ATM machine/website of a partner bank to complete a transaction that credits their mobile prepaid account while debiting their bank account
- ♦ By using a POS terminal of a partner institution that collects the money (via a separate cash, cheque or credit card based transaction) and then credits the mobile prepaid account identified by the subscriber.

 These change requests will be developed on PreTUPS 5.1, these change requests can not be implemented on PreTUPS5.0.

 Comviva has already submitted commercial for version upgrade and we are considering that at the time of implementation of these change requests PreTUPS version will be already upgraded to 5.1.

1.2 Abbreviations and Conventions

Abbreviations	Description
O	Optional (Tag mandatory value optional)
M	Mandatory
HTTP	Hyper Text Transfer Protocol
N	Numeric
A	Alpha Numeric
D	Date
C	Characters.
PreTUPS	Pre Paid TopUp System

Transaction Status	Description
200	Success

Field Type: <Type of Field> (<Maximum length of field>) for e.g.

A (10) means the field has alphanumeric value and the maximum length of field can be 10.

External System TopUp (POS)

This section provides a functional overview of the Top-UP process through external systems.

- Subscriber top-up their mobile account through external system i.e. POS terminal etc

2.1 External System TopUp (POS)

There are two possibilities to implement the External system (POS) interface -

Case 1- PreTUPS receive request for POS server (centralized system), i.e. each POS terminal send request to POS servers and POS servers send request to PreTUPS.

- POS server will be created as Channel user in PreTUPS.
- Channel admin allocate airtime to Pos servers,
- POS do transactions from POS servers account.
- In this case less operational effort but reporting in PreTUPS will be available only at POS servers level. POS level reports will be available on the POS server.

Case 2- Each POS terminal sending request to PreTUPS via POS server.

- Each POS terminal will be created as channel user in PreTUPS.
- Channel admin allocate airtime to each POS terminal.
- POS terminal will do transactions.
- In this case more operational effort but reporting in PreTUPS will be available at each POS level.

 New distribution channel e.g. POS Terminals (External System) in PreTUPS will be added.

 Comviva recommendation is case 1.

2.2 External System API

This PreTUPS API allows the other external systems (POS terminals etc.) to interface with PreTUPS to use the C2S Transfer (recharge, postpaid bill payment, EVD) services.

2.2.1 C2S Transfer (Customer Recharge)

External system will send Customer Recharge request to PreTUPS in the following format:

XML format:

```
<?xml version="1.0"?>
<COMMAND>
  <TYPE>EXRCTRFREQ</TYPE>
  <DATE><Date and time ></DATE>
```

```

<EXTNWCODE><Network External Code></EXTNWCODE>
<MSISDN><Retailer MSISDN></MSISDN>
<PIN><123456></PIN>
<LOGINID><Channel user Login ID></LOGINID>
<PASSWORD><Channel User Login Password></PASSWORD>
<EXTCODE><Channel user unique External code></EXTCODE>
<EXTREFNUM><Unique Reference number in the external
system></EXTREFNUM>
<MSISDN2><Payee MSISDN></MSISDN2>
<AMOUNT><Amount></AMOUNT>
<LANGUAGE1><Retailer Language></LANGUAGE1>
<LANGUAGE2><Payee Language></LANGUAGE2>
<SELECTOR><Selector></SELECTOR>
</COMMAND>

```

Field Details

TAG	Fields	Remarks	Example	Field Type	Optional/ Mandatory
TYPE	Request type	Request Type, should be sent with each request – fixed	EXRCTRFR EQ	A (20)	M
DATE	Date and time	Date and time on which request was sent by external system, HH are in 24 Hour Format	DD/MM/YYYY HH24:MI:SS	D (20)	O (Tag is mandatory)
EXTNWCODE	Network code	Network code of the Channel User defined in PreTUPS as External Network code	MO	A (2)	M
MSISDN	Channel user/Subscriber MSISDN	All MSISDN should be in national dial format i.e. with out country code.	9942222	N (15)	O (Tag is mandatory)
PIN	Channel user/Subscriber PIN	PIN of the user	123	A (10)	O (Tag is mandatory)
LOGINID	Login ID	Login ID of the Channel user	Mo_cce	A (20)	O
PASSWORD	Password	Password of the Channel user	2468	A (10)	O (Tag is mandatory)
EXTCODE	External code of the channel user	Unique external code of the channel user defined in PreTUPS.	123	A (10)	O (Tag is mandatory)
EXTREFNUM	External Reference number	Unique Reference number in the external system. PreTUPS will not check uniqueness	12345	A (20)	O (Tag is mandatory)

TAG	Fields	Remarks	Example	Field Type	Optional/ Mandatory
Note: Between MSISDN, LOGINID and EXTCODE value of one of them must be present, either MSISDN, LOGINID or EXTCODE. All of them can also be present in request					
MSISDN2	Payee MSISDN	All MSISDN should be in national dial format i.e. with out country code.	9942222	N (15)	M
AMOUNT	<Amount>	Amount to be recharge May be in lowest denomination i.e. in cents, \$1 =100 cents	50055 i.e. 500.55	N (20)	M
LANGUAGE1	<Retailer Language>	Numeric only, Retailer Language Code This code must be defined in PreTUPS system.	0	A (10)	O (Tag is mandatory) Default value 0
LANGUAGE2	< Payee Language>	Numeric only, Payee Language Code This code must be defined in PreTUPS system.	0	A(10)	O (Tag is mandatory) Default value 0
SELECTOR	<Selector>	Selector should be numeric 1 – CVG 2- C 3- VG	1	A(10)	M Default value 1

User/Sender validation login

- If MSISDN value is available then user will be validated on the basis of the MSISDN and PIN
 - If User Authentication required is false (means request is coming from authenticated POS servers and POS server has already done validation of POS terminal) then transactions will be allowed without checking PIN/Password.
- If EXTCODE value is available then user will be validated on the basis of the EXTCODE
- If LOGINID tag value is available then user will be validated on the basis of the LOGINID and PASSWORD.
- If values of more than one tag is present then user will be validated depending on those values e.g. (MSISDN and EXT code both present in request then user will be validated on the basis of the both)
- If all the three are available then all will be validated. Between all of them any one would be mandatory.
- Once the sender is validated, then the appropriate amount would be debited from its account.

Response

PreTUPS will send following response (acknowledgement) to External system for customer recharge request:

XML format:

```
<?xml version="1.0"?>
<COMMAND>
  <TYPE>EXRCTRFPRESP</TYPE>
  <TXNSTATUS><Transaction Status></TXNSTATUS >
  <DATE><Date and time></DATE>
  <EXTREFNUM><Unique Reference number in the external system></EXTREFNUM>
  <TXNID><PreTUPS Transaction ID></TXNID>
  <MESSAGE><Transaction Message></MESSAGE>
</COMMAND>
```

Field Details

TAG	Fields	Remarks	Example	Field Type	Optional/ Mandatory
TYPE	Response type	Response Type	EXRCTRFR ESP	A (20)	M
TXNSTATUS	Transaction Status	Status of the request - Transaction Status= 200 means Success, - Transaction Status Other than 200 means failed	200	N (10)	M
DATE	Date and time	Date and time on which response was sent from PreTUPS. HH are in 24 Hour format	DD/MM/YYYY HH24:MI:SS	D (20)	M
EXTREFNUM	External Reference number	Reference number that was passed by the external system	12345	A (20)	O (Tag is mandatory)
TXNID	<Transaction ID>	PreTUPS Transaction ID for the Customer Recharge Transaction	R080912.12 12.1234	20	M
MESSAGE	Message that will given in response	Message		A (500)	O (Tag is mandatory)

-  If the TXNSTATUS is not 200 then the user would be shown the error message (MESSAGE tag) returned in the response.
-  If due to some reason external system does not received response of the above request then External system can send the recharge status request to PreTUPS for checking the final status of the transaction by sending its External Reference number.
-  If transaction status is 200 then PreTUPS will send Previous and post balance in the message tag.

2.2.2 C2S Transfer (Bill Payment)

External system will send Customer Bill Payment request to PreTUPS in the following format:

XML format:

```
<?xml version="1.0"?>
<COMMAND>
  <TYPE>EXPPBREQ</TYPE>
  <DATE><Date and time></DATE>
  <EXTNWCODE><Network External Code></EXTNWCODE>
  <MSISDN><Retailer MSISDN></MSISDN>
  <PIN><123456></PIN>
  <LOGINID><Channel user Login ID></LOGINID>
  <PASSWORD><Channel User Login Password></PASSWORD>
  <EXTCODE><Channel user unique External code></EXTCODE>
  <EXTREFNUM><Unique Reference number in the external
system></EXTREFNUM>
  <MSISDN2><Payee MSISDN></MSISDN2>
  <AMOUNT><Amount></AMOUNT>
  <LANGUAGE1><Retailer Language></LANGUAGE1>
  <LANGUAGE2><Payee Language></LANGUAGE2>
  <SELECTOR><Selector></SELECTOR>
</COMMAND>
```

Field Details

TAG	Fields	Remarks	Example	Field Type	Optional/Mandatory
TYPE	Request type	Request Type, should be sent with each request – fixed	EXPPBREQ	A (20)	M
DATE	Date and time	Date and time on which request was sent by external system, HH are in 24 Hour Format	DD/MM/YYYY HH24:MI:SS	D (20)	O (Tag is mandatory)
EXTNWCODE	Network code	Network code of the Channel User defined in PreTUPS as External Network code	MO	A (2)	M

TAG	Fields	Remarks	Example	Field Type	Optional/ Mandatory
MSISDN	Channel user/Subscriber MSISDN	All MSISDN should be in national dial format i.e. with out country code.	9942222	N (15)	O (Tag is mandatory)
PIN	Channel user/Subscriber PIN	PIN of the user	123	A (10)	O (Tag is mandatory)
LOGINID	Login ID	Login ID of the Channel user	Mo_cce	A (20)	O
PASSWORD	Password	Password of the Channel user	2468	A (10)	O (Tag is mandatory)
EXTCODE	External code of the channel user	Unique external code of the channel user defined in PreTUPS.	123	A (10)	O (Tag is mandatory)
EXTREFNUM	External Reference number	Unique Reference number in the external system. PreTUPS will not check uniqueness	12345	A (20)	O (Tag is mandatory)
Note: Between MSISDN, LOGINID and EXTCODE value of one of them must be present, MSISDN, LOGINID or EXTCODE. All of them can also be present in request					
MSISDN2	Payee MSISDN	All MSISDN should be in national dial format i.e. with out country code.	9942222	N (15)	M
AMOUNT	<Amount>	Amount to be recharge May be in lowest denomination i.e. in cents, \$1 =100 cents	50055 i.e. 500.55	N (20)	M
LANGUAGE1	<Retailer Language>	Numeric only, Retailer Language Code This code must be defined in PreTUPS system.	0	A (10)	O (Tag is mandatory) Default value 0
LANGUAGE2	< Payee Language>	Numeric only, Payee Language Code This code must be defined in PreTUPS system.	0	A(10)	O (Tag is mandatory) Default value 0
SELECTOR	<Selector>	Selector should be numeric 2- C	2	A(10)	M Default value 2

User/Sender validation login

- If MSISDN value is available then user will be validated on the basis of the MSISDN and PIN
- If EXTCODE value is available then user will be validated on the basis of the EXTCODE
- If LOGINID tag value is available then user will be validated on the basis of the LOGINID and PASSWORD.
- If values of more than one tag is present then user will be validated depending on those values e.g. (MSISDN and EXT code both present in request then user will be validated on the basis of the both)

- If all the three are available then all will be validated. Between all of them any one would be mandatory.
- Once the sender is validated, then the appropriate amount would be debited from its account.

Response

PreTUPS will send following response (acknowledgement) to External system for bill payment request:

XML format:

```
<?xml version="1.0"?>
<COMMAND>
  <TYPE>EXPPBRESP</TYPE>
  <TXNSTATUS><Transaction Status></TXNSTATUS>
  <DATE><Date and time></DATE>
  <EXTREFNUM><Unique Reference number in the external system></EXTREFNUM>
  <TXNID><PreTUPS Transaction ID></TXNID>
  <MESSAGE><Transaction Message></MESSAGE>
</COMMAND>
```

Field Details

TAG	Fields	Remarks	Example	Field Type	Optional/ Mandatory
TYPE	Response type	Response Type	EXPPBRESP	A (20)	M
TXNSTATUS	Transaction Status	Status of the request - Transaction Status= 200 means Success, - Transaction Status Other than 200 means failed	200	N (10)	M
DATE	Date and time	Date and time on which response was sent from PreTUPS. HH are in 24 Hour format	DD/MM/YYYY HH24:MI:SS	D (20)	M
EXTREFNUM	External Reference number	Reference number that was passed by the external system	12345	A (20)	O (Tag is mandatory)
TXNID	<Transaction ID>	Pretups Transaction ID for the Customer Recharge Transaction	R080912.12 12.1234	20	M
MESSAGE	Message that will given in response	Message		A (500)	O (Tag is mandatory)

-  If the TXNSTATUS is not 200 then the user would be shown the error message (MESSAGE tag) returned in the response.
-  If due to some reason external system does not received response of the above request then External transaction server can send the recharge status request to PreTUPS for checking the final status of the transaction by sending its External Reference number.
-  If transaction status is 200 then PreTUPS will send Previous and pos balance in the message tag.

2.2.3 C2S Transfer (EVD – Electronic Voucher Distribution)

External system will send Electronic Voucher Distribution request to PreTUPS in the following format:

XML format:

```
<?xml version="1.0"?>
<COMMAND>
  <TYPE>EXEVDREQ</TYPE>
  <DATE><Date and time></DATE>
  <EXTNWCODE><Network External Code></EXTNWCODE>
  <MSISDN><Retailer MSISDN></MSISDN>
  <PIN><123456></PIN>
  <LOGINID><Channel user Login ID></LOGINID>
  <PASSWORD><Channel User Login Password></PASSWORD>
  <EXTCODE><Channel user unique External code></EXTCODE>
  <EXTREFNUM><Unique Reference number in the external
system></EXTREFNUM>
  <MSISDN2><Payee MSISDN></MSISDN2>
  <AMOUNT><Amount></AMOUNT>
  <LANGUAGE1><Retailer Language></LANGUAGE1>
  <LANGUAGE2><Payee Language></LANGUAGE2>
  <SELECTOR><Selector></SELECTOR>
</COMMAND>
```

Field Details

TAG	Fields	Remarks	Example	Field Type	Optional/ Mandatory
TYPE	Request type	Request Type, should be sent with each request – fixed	EXEVDREQ	A (20)	M
DATE	Date and time	Date and time on which request was sent by external system, HH are in 24 Hour Format	DD/MM/YYYY HH24:MI:SS	D (20)	O (Tag is mandatory)
EXTNWCODE	Network code	Network code of the Channel User defined in PreTUPS as External	MO	A (2)	M

TAG	Fields	Remarks	Example	Field Type	Optional/ Mandatory
		Network code			
MSISDN	Channel user/Subscriber MSISDN	All MSISDN should be in national dial format i.e. with out country code.	9942222	N (15)	O (Tag is mandatory)
PIN	Channel user/Subscriber PIN	PIN of the user	123	A (10)	O (Tag is mandatory)
LOGINID	Login ID	Login ID of the Channel user	Mo_cce	A (20)	O (Tag is mandatory)
PASSWORD	Password	Password of the Channel user	2468	A (10)	O (Tag is mandatory)
EXTCODE	External code of the channel user	Unique external code of the channel user defined in PreTUPS.	123	A (10)	O (Tag is mandatory)
EXTREFNUM	External Reference number	Unique Reference number in the external system. PreTUPS will not check uniqueness	12345	A (20)	O (Tag is mandatory)
Note: Between MSISDN, LOGINID and EXTCODE value of one of them must be present, either MSISDN, LOGINID or EXTCODE. All of them can also be present in request					
MSISDN2	Payee MSISDN	All MSISDN should be in national dial format i.e. with out country code.	9942222	N (15)	M
AMOUNT	<Amount>	Voucher Amount May be in lowest denomination i.e. in cents, \$1 =100 cents	50055 i.e. 500.55	N (20)	M
LANGUAGE1	<Retailer Language>	Numeric only, Retailer Language Code This code must be defined in PreTUPS system.	0	A (10)	O (Tag is mandatory) Default value 0
LANGUAGE2	< Payee Language>	Numeric only, Payee Language Code This code must be defined in PreTUPS system.	0	A(10)	O (Tag is mandatory) Default value 0
SELECTOR	<Selector>	Selector should be numeric 1 – CVG	1	A(10)	M Default value 1

User/Sender validation login

- If MSISDN value is available then user will be validated on the basis of the MSISDN and PIN
- If EXTCODE value is available then user will be validated on the basis of the EXTCODE
- If LOGINID tag value is available then user will be validated on the basis of the LOGINID and PASSWORD.
- If values of more than one tag is present then user will be validated depending on those values e.g. (MSISDN and EXT code both present in request then user will be validated on the basis of the both)
- If all the three are available then all will be validated. Between all of them any one would be mandatory.
- Once the sender is validated, then the appropriate amount would be debited from its account.

Response

PreTUPS will send following response (acknowledgement) to External system for Electronic Voucher Distribution request:

XML format:

```
<?xml version="1.0"?>
<COMMAND>
  <TYPE>EXEVDRESP</TYPE>
  <TXNSTATUS><Transaction Status></TXNSTATUS>
  <DATE><Date and time></DATE>
  <EXTREFNUM><Unique Reference number in the external system></EXTREFNUM>
  <TXNID><PreTUPS Transaction ID></TXNID>
  <MESSAGE><Transaction Message></MESSAGE>
</COMMAND>
```

Field Details

TAG	Fields	Remarks	Example	Field Type	Optional/ Mandatory
TYPE	Response type	Response Type	EXEVDRESP	A (20)	M
TXNSTATUS	Transaction Status	Status of the request ■ Transaction Status= 200 means Success, ■ Transaction Status Other than 200 means failed	200	N (10)	M
DATE	Date and time	Date and time on which response was sent from PreTUPS. HH are in 24	DD/MM/YYYY HH24:MI:SS	D (20)	M

TAG	Fields	Remarks	Example	Field Type	Optional/ Mandatory
		Hour format			
EXTREFNUM	External Reference number	Reference number that was passed by the external system	12345	A (20)	O (Tag is mandatory)
TXNID	<Transaction ID>	PreTUPS Transaction ID for the Customer Recharge Transaction	R080912.12 12.1234	20	M
MESSAGE	Message that will given in response	Message		A (500)	O (Tag is mandatory)

-  If the TXNSTATUS is not 200 then the user would be shown the error message (MESSAGE tag) returned in the response.
-  If due to some reason external system does not received response of the above request then External transaction server can send the recharge status request to PreTUPS for checking the final status of the transaction by sending its External Reference number.
-  If transaction status is 200 then PreTUPS will send Previous and post balance in the message tag.

2.2.4 C2S Transfer Status (Recharge, Postpaid Bill payment, EVD)

External system will send C2S transfer status request to PreTUPS in the following format:
The external system to check the final status of the C2S Transfer requests (recharge, postpaid bill payment, voucher distribution) will use this API

XML format

```
<?xml version="1.0"?>
<!DOCTYPE COMMAND PUBLIC "-//Ocam//DTD XML Command 1.0//EN"
"xml/command.dtd">
<COMMAND>
<TYPE>EXRCSTATREQ</TYPE>
<DATE><Date and time></DATE>
<EXTNWCODE><Network External Code></EXTNWCODE>
<MSISDN><Retailer MSISDN></MSISDN>
<PIN><Pin></PIN>
<LOGINID><Channel user Login ID></LOGINID>
<PASSWORD><Channel User Login Password></PASSWORD>
<EXTCODE><Channel user unique External code></EXTCODE>
<EXTREFNUM><Unique Reference number of previous request></EXTREFNUM>
<TXNID><PreTUPS TXN ID></TXNID>
<LANGUAGE1><Retailer Language></LANGUAGE1>
```

</COMMAND>

Field Details

TAG	Fields	Remarks	Example	Field Type	Optional/ Mandatory
TYPE	Request type	Request Type, should be sent with each request - fixed	EXRCSTATREQ	A (20)	M
DATE	Date and time	Date and time on which request generated by external transaction server, HH are in 24 Hour Format	DD/MM/YYYY HH24:MI:SS	D (20)	O (Tag is mandatory)
EXTNWCODE	Network code	Network code of the Channel User defined in PreTUPS as External Network code	MO	A (2)	M
MSISDN	Channel user/Subscriber MSISDN	All MSISDN should be in national dial format i.e. with out country code.	9942222	N (15)	O (Tag is mandatory)
PIN	Channel user/Subscriber PIN	PIN of the user	123	A (10)	O
LOGINID	Login ID	Login ID of the Channel user	Mo_cce	A (20)	O (Tag is mandatory)
PASSWORD	Password	Password of the Channel user	2468	A (10)	O (Tag is mandatory)
EXTCODE	External code of the channel user	Unique external code of the channel user defined in PreTUPS.	123	A (10)	O (Tag is mandatory)
EXTREFNUM	External Reference number	Unique Reference number in the sent with transaction request	12345	A (20)	M
Note: Between MSISDN, LOGINID and EXTCODE one of them must be present, either MSISDN, LOGINID or EXTCODE. All of them can also be present in request					
TXNID	<Transaction ID>	Pretups Transaction ID for the Customer Recharge Transaction	R080912.1212.1 234	20	M (Tag is mandatory)

TAG	Fields	Remarks	Example	Field Type	Optional/ Mandatory
LANGUAGE1	<Retailer Language>	Numeric only, Retailer Language Code This code must be defined in PreTUPS system.	0	1	O (Tag is mandatory) Default value 0

Response

PreTUPS will send the following response (acknowledgement) to External system for C2S transfer status request.

XML format

```
<?xml version="1.0"?>
<COMMAND>
<TYPE>EXRCSTATRESP</TYPE>
<TXNSTATUS><Transaction Status></TXNSTATUS>
<DATE><Date and time></DATE>
<EXTREFNUM><Reference number of transaction request></EXTREFNUM>
<TXNID><PreTUPS Transaction ID></TXNID>
<REQSTATUS><Request Status></REQSTATUS>
<MESSAGE><Message></MESSAGE>
</COMMAND>
```

Field Details

TAG	Fields	Remarks	Example	Field Type	Optional/ Mandatory
TYPE	Response type	Response Type	EXRCSTATRESP	A (20)	M
TXNSTATUS	Transaction Status	Status of the request - Transaction Status= 200 means Success, - Transaction Status Other than 200 means failed	200	N (10)	M
DATE	Date and time	Date and time on which PreTUPS sent response. HH are in 24 Hour format	DD/MM/YYYY HH24:MI:SS	D (20)	M
EXTREFNUM	External Reference number	Reference number that was passed by the external transaction server	12345	A (20)	O (Tag is mandatory)

TAG	Fields	Remarks	Example	Field Type	Optional/ Mandatory
TXNID	<Transaction ID>	PreTUPS Transaction ID for the Customer Recharge Transaction	R080912.1212.1234	20	M
REQSTATUS	Transaction ID Status of the One queried	206= Fail 200=Success 205=Under Process 250=Ambiguous	206	A(20)	M
MESSAGE	Message that will given in response	Message	Transaction number DL/05/000000015 to recharge 100 INR to 9810012345 is successful. Receiver transferred value 99 INR	A (500)	O (Tag is mandatory)

2.2.5 Balance Enquiry

External system will send channel user balance enquiry to PreTUPS in the following format:

XML format

```
<?xml version="1.0"?>
<!DOCTYPE COMMAND PUBLIC "-//Ocam//DTD XML Command 1.0//EN"
"xml/command.dtd">
<COMMAND>
<TYPE>EXUSRBALREQ</TYPE>
<DATE><Date and time></DATE>
<EXTNWCODE><Network External Code></EXTNWCODE>
<MSISDN><Retailer MSISDN></MSISDN>
<PIN><Pin></PIN>
<LOGINID><Channel user Login ID></LOGINID>
<PASSWORD><Channel User Login Password></PASSWORD>
<EXTCODE>< Channel user unique External code></EXTCODE>
<EXTREFNUM><Reference number of transaction request></EXTREFNUM>
</COMMAND>
```

Field Details

TAG	Fields	Remarks	Example	Field Type	Optional/ Mandatory
TYPE	Request type	Request Type,	EXUSRBALREQ	A (20)	M

TAG	Fields	Remarks	Example	Field Type	Optional/ Mandatory
		should be sent with each request – fixed			
DATE	Date and time	Date and time on which request generated by external transaction server, HH are in 24 Hour Format	DD/MM/YYYY HH24:MI:SS	D (20)	O (Tag is mandatory)
EXTNWCODE	Network code	Network code of the Channel User defined in PreTUPS as External Network code	AK	A (2)	M
MSISDN	Channel user/Subscriber MSISDN	All MSISDN should be in national dial format i.e. with out country code.	9942222	N (15)	O (Tag is mandatory)
PIN	Channel user/Subscriber PIN	PIN of the user	123	A (10)	O
LOGINID	Login ID	Login ID of the Channel user	Mo_cce	A (20)	O (Tag is mandatory)
PASSWORD	Password	Password of the Channel user	2468	A (10)	O (Tag is mandatory)
EXTCODE	External code of the channel user	Unique external code of the channel user defined in PreTUPS.	123	A (10)	O (Tag is mandatory)
EXTREFNUM	External Reference number	Reference number of external system.	12345	A (20)	O (Tag is mandatory)

Note: Between MSISDN, LOGINID and EXTCODE one of them must be present, either MSISDN, LOGINID or EXTCODE. All of them can also be present in request.

Response

PreTUPS will send the following response (acknowledgement) to External system for the channel user balance enquiry request.

XML format

```
<?xml version="1.0"?>
<COMMAND>
<TYPE>EXUSRBALRESP</TYPE>
<TXNSTATUS><Transaction Status></TXNSTATUS>
```

```

<DATE><Date and time></DATE>
<EXTREFNUM><Reference number of transaction
request></EXTREFNUM>
<RECORD>
  <PRODUCTCODE><Product Code></PRODUCTCODE>
  <PRODUCTSHORTNAME><Short code of
product></PRODUCTSHORTNAME>
  <BALANCE><Balance of product></BALANCE>
</RECORD>
<RECORD>
  <PRODUCTCODE><Product Code></PRODUCTCODE>
  <PRODUCTSHORTNAME><Short code of
product></PRODUCTSHORTNAME>
  <BALANCE><Balance of product></BALANCE>
</RECORD>
<MESSAGE><Message></MESSAGE>
</COMMAND>

```

Field Details

TAG	Fields	Remarks	Example	Field Type	Optional/ Mandatory
TYPE	Response type	Response Type	EXUSRBALRES P	A (20)	M
TXNSTATUS	Transaction Status	Status of the request - Transaction Status= 200 means Success, - Transaction Status Other than 200 means failed	200	N (10)	M
DATE	Date and time	Date and time on which PreTUPS sent response. HH are in 24 Hour format	DD/MM/YYYY HH24:MI:SS	D (20)	M
EXTREFNUM	External Reference number	Reference number that was passed by the external transaction server	12345	A (20)	O (Tag is mandatory)
MESSAGE	Message that will given in response	Message		A (500)	O (Tag is mandatory)
RECORD Tag to identify a single record. Will be repeated in case of the multiple products for a channel user. In case of the single product then only once sub tags will be present.					
PRODUCTCODE	Product code of the product	Product code of the product		N (10)	M
PRODUCTSHORT NAME	Short name of the product	Short name of the product		A (15)	M

TAG	Fields	Remarks	Example	Field Type	Optional/ Mandatory
BALANCE	Balance of the product	Balance of the product in higher denomination of local currency e.g. \$		N (20,2)	M

2.3 Communication Protocol

The communication between External system and PreTUPS is XML over HTTP. The External system makes the HTTP connection with PreTUPS and sends the request (Post) content as XML. Response of each request is also sent as XML.

2.3.1 Authentication

To authenticate the External system, PreTUPS will define a Gateway for External system. PreTUPS Engineer give the defined gateway information details to External system users. External system will send this gateway information in header parameter for each request.

Note: - There are two level of authentication for External API, one will come in header parameter explains in this section and another will come along with the request in post method.

Details of authentication information are as follows

- LOGIN
- PASSWORD
- REQUEST_GATEWAY_CODE
- REQUEST_GATEWAY_TYPE
- SERVICE_PORT
- SOURCE_TYPE

Authentication information

LOGIN=<Login>&PASSWORD=<Encrypted Password>&REQUEST_GATEWAY_CODE=<Request Gateway Code>&REQUEST_GATEWAY_TYPE=<Request Gateway Type>&SERVICE_PORT=<Service Port>&SOURCE_TYPE=<Source Type>

Example

LOGIN=test&PASSWORD=znclfhdy35ed&REQUEST_GATEWAY_CODE=EXTGW001&REQUEST_GATEWAY_TYPE=EXTGW&SERVICE_PORT=190&SOURCE_TYPE=EXT



Details of above-mentioned field are described in Connection procedure section.

2.3.2 Connection Procedure

The External system calls the specific URL (provided by Comviva) to make the HTTP Connection with PreTUPS. The details of HTTP Header and URL are mentioned below.

HTTP Header Information

POST <URL> HTTP

Content-type: xml

Connection: close

Authorization: LOGIN=<Login>&PASSWORD=<Encrypted Password>&REQUEST_GATEWAY_CODE=<Request Gateway Code>&REQUEST_GATEWAY_TYPE=<Request Gateway Type>&SERVICE_PORT=<Service Port>&SOURCE_TYPE=<Source Type>

URL

http://<PreTUPSHost>/C2SReceiver?

Example

URL for Recharge Service

http://<pretupshost>/C2SReceiver?

Field Details

Fields	Description	Example	Optional/ Mandatory	Field Type
PreTUPSHost	The IP, port and host name. The IP and Port on which PreTUPS application is running	172.16.1.109:5555/pr etups/	M	N.A.
LOGIN	Login Id provided to External system by PreTUPS to authenticate External System	Test	M	A (10)
PASSWORD	Encrypted Password provided to External system by PreTUPS to authenticate External system	Test	M	A (10)
REQUEST_GATEWAY_CODE	Request Gateway Code provided to External system by PreTUPS to identify request interface	EXT001	M	A (10)
REQUEST_GATEWAY_TYPE	Request Gateway Type provided to External system by PreTUPS to identify request interface type.	EXTSYS	M	A (10)
SERVICE_PORT	Service port provided to External system by PreTUPS to identify service port	126	M	N (10)
SOURCE_TYPE	Source Type of the request	EXTSYS	M	A (10)

2.3.3 Security

The interface between the External system & PreTUPS is expected to carry requests for credit a subscriber's prepaid account. This requires a high security environment. Therefore,

the physical interface between the PreTUPS & External system is expected to be over private leased circuits/Virtual Private Network Tunnels/ISDN Dialup. It is not expected that this interchange will occur over the public Internet.

Appendix 1 - Error Codes

Status of the request

- Transaction Status= 200 means Success,
- Transaction Status Other than 200 means failed

Note: -

- If the TXNSTATUS is not 200 then the user would be shown the error message (MESSAGE tag) returned in the response.
- If due to some reason external system does not receive response of the above request then External transaction server can send the recharge status request to PreTUPS for checking the final status of the transaction by sending its External Reference number.

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